

Chapter 1

Introduction and Research Design

1.1 Motivation: Equity Market Concentration and Systemic Fragility

The S&P 500 plays several roles in today’s financial system: it is the benchmark for institutional performance evaluation, the core building block of passive investment products, a key input for risk models, and the underlying asset for the most liquid index options markets. As such, changes in its internal structure can have consequences that extend beyond portfolio composition and into the domain of market-wide fragility. Over the past decade, it has been repeatedly documented that the S&P 500 has become markedly more top-heavy, with a growing share of index weight concentrated in a handful of mega-cap firms. According to S&P Global, concentration in the S&P 500 has risen to the point where roughly two-fifths of the index’s value is now tied to just ten firms, a configuration comparable only to episodes in the 1960s [7]. This development has motivated index providers to launch alternative benchmarks such as a dedicated “top-ten” index that isolates the largest constituents within the S&P 500 [11] and equal-weight variants of the index that redistribute portfolio weight away from the mega caps and restore a more balanced exposure to the broader opportunity set [10]. At a fundamental level, this dissertation asks whether what we routinely call “the market” still behaves like a broadly diversified portfolio or has effectively become a concentrated exposure to a small set of firms and risk factors.

The simplest way to describe concentration is in cross-sectional terms: in any cap-weighted index, larger firms carry higher weights, so if a few companies grow markedly faster than the rest, the index naturally becomes more top-heavy. Common summary statistics, such as the cumulative weight of the largest stocks (“top-N” weight shares), a Herfindahl–Hirschman Index (HHI), or a Gini coefficient of index weights, are used in index research to quantify concentration levels and compare them across time and within sectors [9]. While such weight-based statistics are useful as a starting point, they are not sufficient to characterize the economic implications of concentration, as they remain silent about the mechanism of risk aggregation. The reason is that the market index is not merely a collection of weights: it is a risk aggregator, and systemic fragility is driven by how

shocks propagate into the aggregate distribution, particularly in the left tail and during episodes when correlations spikes and normal diversification breaks down. As an ever larger fraction of index returns and volatility is driven by a small set of dominant firms, the apparent breadth of the benchmark can mask a more fragile form of diversification: upside participation looks easy in rising markets, but disappointments in one or two key names can translate into disproportionately severe index level drawdowns. Put differently, high concentration can amplify systemic fragility in ways that a static weight metric alone would fail to reveal.

For these reasons, this dissertation treats concentration as more than a mechanical by-product of index construction. The guiding idea is to view concentration as a structural state of the equity market, in which aggregate outcomes become disproportionately sensitive to the shocks, exposures, and tail risks of a narrow set of dominant firms. This state-based perspective closely parallels the granular hypothesis, a classical macro-finance idea that argues that when the size distribution is sufficiently skewed (i.e., few firms account for a large share of the economy), idiosyncratic shocks do not wash out in the aggregate and can instead meaningfully contribute to aggregate fluctuations [8]. In the context of a concentrated equity index, the granular mechanism suggests that what is often labelled “idiosyncratic” at the firm level may become “systemic” at the index level because the index may no longer behave like the average of many small, roughly similar exposures [8].

One might object that top-heaviness is just a diversification issue: if the effective breadth of the index shrinks, idiosyncratic risk is less diluted. While correct, that framing is incomplete for two reasons. First, the welfare-relevant dimension of equity risk is not limited to variance, as rational investors care not only about the overall level of volatility but also about the shape of the return distribution: the magnitude of large losses, the persistence of drawdowns, and the frequency with which the market enters correlation regimes in which diversification becomes ineffective, i.e., regimes characterized by state-dependent correlations that undermine diversification in stress periods. Second, such adverse states are not only realized ex post, but they are also priced ex ante, as the market prices adverse states directly through derivatives: equity index options embed forward-looking information about the distribution of market outcomes, especially crash scenarios, and thereby reveal how much investors are willing to pay for insurance against market-wide downturns [5]. In practice, indices such as the VIX aggregate information from SPX options into a measure of expected short horizon volatility, illustrating how option markets encode investors’ willingness to pay for protection against market wide downturns [6]. Option based studies show that the variance risk premium (VRP), i.e., the difference between risk neutral and expected realized variance, forecasts future equity returns and is strongly linked to downside risk [4]. Andersen et al. (2015) show that a separate “left-tail” factor in the option-implied density is required to explain market crashes beyond what volatility alone can predict [3]. This motivates the core theme of the dissertation: equity market concentration should be analyzed as a potential driver of fragility in both in the physical return distribution and in the risk-neutral pricing of extreme events, i.e., tail states.

In short, the rise in concentration is not only a story about benchmark construction. It raises the possibility that the “market portfolio” itself may have shifted toward a regime in which aggregate risk is heavily shaped by a small cluster of firms and the dependence structure surrounding them. These are precisely the ingredients that lie at the core of systemic-risk analysis, as captured by measures such as CoVar, MES, and SRISK, which

formalise tail dependence and conditional capital shortfalls at the system level [2] [1]. Yet existing systemic risk measures typically focus on financial institutions and balance sheet interlinkages, rather than on how changes in index composition reshape aggregate equity risk. This motivates the need to go beyond weight-based concentration metrics and to reconceptualize concentration as a structural state variable of the equity market.

1.2 From Concentration as a Cross-Sectional Fact to a Structural State Variable

Bibliography

- [1] Viral V. Acharya et al. “Measuring Systemic Risk”. In: *The Review of Financial Studies* 30.1 (2017), pp. 2–47. DOI: [10.1093/rfs/hhw088](https://doi.org/10.1093/rfs/hhw088).
- [2] Tobias Adrian and Markus K. Brunnermeier. “CoVaR”. In: *American Economic Review* 106.7 (2016), pp. 1705–1741. DOI: [10.1257/aer.20120555](https://doi.org/10.1257/aer.20120555).
- [3] Torben G. Andersen, Nicola Fusari, and Viktor Todorov. “The Risk Premia Embedded in Index Options”. In: *Journal of Financial Economics* 117.3 (2015), pp. 558–584. DOI: [10.1016/j.jfineco.2015.06.005](https://doi.org/10.1016/j.jfineco.2015.06.005).
- [4] Tim Bollerslev, George Tauchen, and Hao Zhou. “Expected Stock Returns and Variance Risk Premia”. In: *The Review of Financial Studies* 22.11 (2009), pp. 4463–4492. DOI: [10.1093/rfs/hhp008](https://doi.org/10.1093/rfs/hhp008).
- [5] Douglas T. Breeden and Robert H. Litzenberger. “Prices of State-Contingent Claims Implicit in Option Prices”. In: *The Journal of Business* 51.4 (1978), pp. 621–651. URL: <https://www.jstor.org/stable/2352653>.
- [6] Cboe Global Markets. *Cboe Volatility Index: White Paper*. Tech. rep. Cboe Global Markets, 2019. URL: https://www.cboe.com/tradable_products/vix/vix_white_paper.pdf.
- [7] Tim Edwards. *In the shadows of giants: Navigating the long-term dynamics of performance and concentration in US equities*. Tech. rep. S&P Global, 2025. URL: <https://www.spglobal.com/en/research-insights/special-reports/look-forward/partner-perspectives/unlocking-potential-ahead-with-vanguard/in-the-shadows-of-giants>.
- [8] Xavier Gabaix. “The Granular Origins of Aggregate Fluctuations”. NBER Working Paper No. 15286. Aug. 2009. DOI: [10.3386/w15286](https://doi.org/10.3386/w15286). URL: <https://www.nber.org/papers/w15286>.
- [9] Anu R. Ganti and Craig J. Lazzara. *Concentration within Sectors and Its Implications for Equal Weighting*. S&P Global. Accessed: May 2026. 2022. URL: <https://www.spglobal.com/spdji/en/indices/equity/sp-500-equal-weight-index/>.
- [10] S&P Dow Jones Indices. *S&P 500 Equal Weight Index*. S&P Global. Accessed: May 2026. URL: <https://www.spglobal.com/spdji/en/indices/equity/sp-500-equal-weight-index/>.
- [11] S&P Dow Jones Indices. *S&P 500 Top 10 Index*. S&P Global. Accessed: May 2026. URL: <https://www.spglobal.com/spdji/en/indices/equity/sp-500-top-10-index/>.